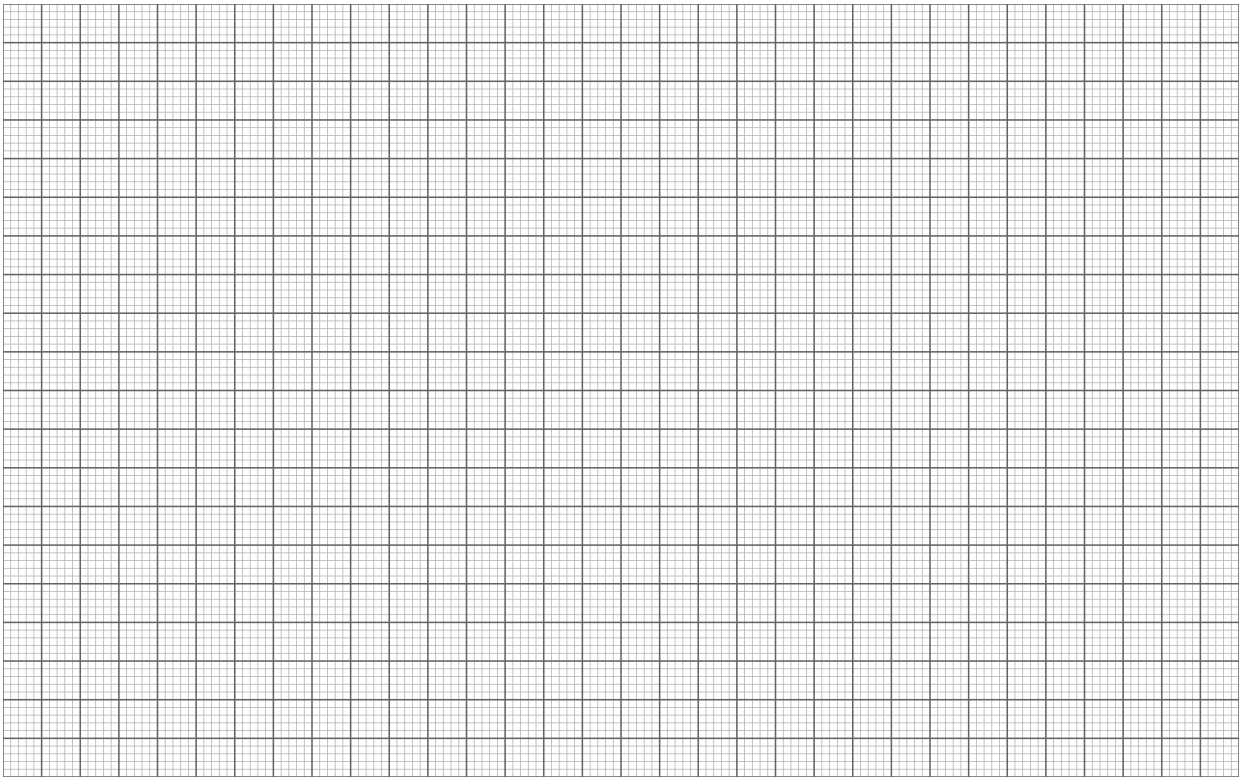
Non-ideal capacitors (10 points)

Part A. Capacitors at room temperature (4.0 points)

A.1 (2.3 pt)
Graph $C_1(U)$ and $C_2(U)$ on the millimeter paper:



Fill the table with the corresponding values:

U	C_1	C_2
$0V$		
$3V$		
$6V$		

$C(U)$ =

Experiment conditions:

Board ID =

T_{room} =

A.2 (0.5 pt)

$U_{\text{max change}}$ =

At capacitor (check): ☐ C1 ☐ C2

A.3 (1.2 pt)

q_1 =

q_2 =

Part B. Calibrating NTC thermistor (1.0 point)

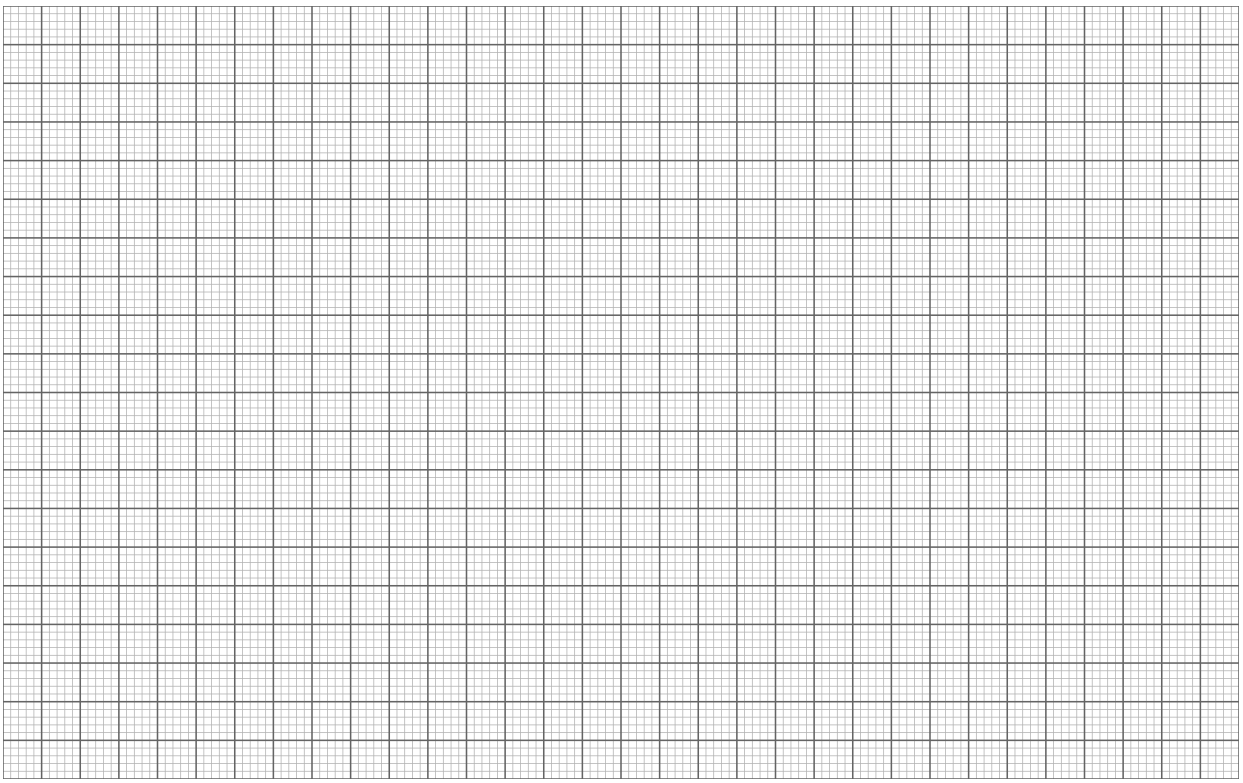
B.1 (1.0 pt)
Formula:

R_0 =

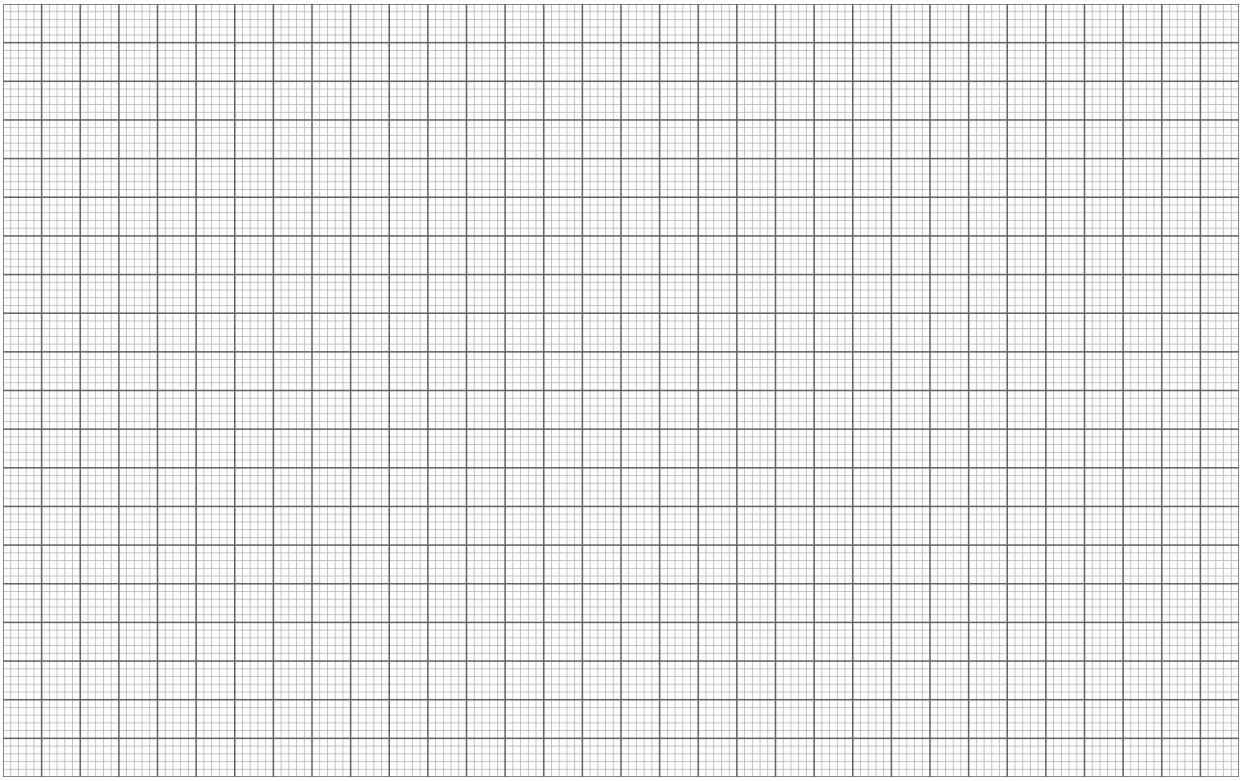
Value:
R_0 = Ω

Part C. Capacitors at different temperatures (3.0 points)

C.1 $(1.3\ \mathrm{pt})$
Graph $C_1(U)$ and $C_2(U)$ on the millimeter paper:



C.2 (0.5 pt)
Graph $C_1(T)$ and $C_2(T)$ on the millimeter paper:



C.3 (1.2 pt)
$C_1(85°\mathrm{C})/C_1(40°\mathrm{C})|_{0\ \mathrm{V}}$ =

$C_1(85°\mathrm{C})/C_1(40°\mathrm{C})|_{6\ \mathrm{V}}$ =

$C_2(85°\mathrm{C})/C_2(40°\mathrm{C})|_{0\ \mathrm{V}}$ =

$C_2(85°\mathrm{C})/C_2(40°\mathrm{C})|_{6\ \mathrm{V}}$ =

Part D. Sources of measurement errors (2.0 points)

D.1 (1.0 pt)

Possible S1 positions: C1, C2

Possible IN connection: +9V, -9V, GND, Free

Initial settings:

S1 position	IN connection

Process:

Step number	S1 position	IN connection	Duration, s	Measured variable

Verification:

Main source of error (check):

☐ Leakage current

☐ Polarization properties of the capacitor's dielectric media

D.2 (1.0 pt)

Possible S1 positions: C1, C2

Possible IN connection: +9V, -9V, GND, Free

Initial settings:

S1 position	IN connection

Process:

Step number	S1 position	IN connection	Duration, s	Measured variable

Verification:

Main source of error (check):

☐ Leakage current

☐ Polarization properties of the capacitor's dielectric media